Edge-QSFL: Quantum-Enhanced Split Federated Learning With Multi-Head Temporal Networks for Edge-Based Intelligent Transportation Security

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***Abstract*—The fast development of Intelligent Transportation Systems (ITS) and the Internet of Vehicles (IoV) has shifted the consumer mobility experience. However, the rapid expansion of such technologies presents serious cybersecurity risks, demanding the implementation of a privacy-preserving, adaptive, and powerful Intrusion Detection System (IDS). Currently, Federated Learning (FL) paradigms in ITS are vulnerable to data poisoning and model-based challenges in diverse applications due to a lack of particular architectures capable of processing temporal data streams. This paper introduces Edge-QSFL, a novel framework that combines quantum-enhanced Split Federated Learning (SFL) and multi-head temporal networks. Our approach extracts features from IoV streaming data using lightweight multi-head Temporal Convolutional Networks (TCN) on edge clients. Subsequently, a quantum-based adaptive aggregation mechanism is implemented by a global server. This mechanism employs dynamic weighting to guarantee that the learning process is resilient to data heterogeneity and poisoning. The architecture also incorporates quantum-enhanced security packages to protect the integrity of models. Extensive analyses of three sample datasets, CICIoV2024, N-BaIoT, and UAVAttackData, indicate almost perfect accuracy scores of 99.42%, 94.82%, and 98.0%, respectively, and show to be more resilient to cyber threats. The results reveal that Edge-QSFL presents a scalable, secure, and high-performance solution, making it suitable for the next generation of ITS networks.**

***Index Terms*—Quantum machine learning, split federated learning, edge intelligence, adaptive intrusion detection, intelligent transportation, quantum cryptography.**

I. Introduction

THE rapid digital transformation of ITS, driven by IoV and connected consumer infrastructure, has allowed for innovative forms of urban mobility and traffic optimization. The advancement of smart vehicles, roadside units, and transportation sensors has led to the development of networked ecosystems that generate valuable mobility and safety data [1], [2]. This transformation could enhance public safety, travel times, and congestion management through real-time data analytics and automated maintenance procedures.

Edge computing has emerged as an important paradigm for efficient and low-latency data processing in ITS. Analyzing vehicle data locally saves response times and bandwidth usage while addressing privacy concerns about transferring sensitive data to clouds. However, due to their distributed and large-scale nature, modern transportation networks are highly vulnerable to sophisticated cyberattacks [3]. Adversarial actors pose a threat to the security of the population if they are able to disrupt vehicle routing algorithms or inflict substantial infrastructure harm by attacking traffic management systems. This requires a strong cybersecurity paradigm with effective defensive capabilities to protect the collaborative operational integrity. In this scenario, the centralized nature of traditional detection systems also encounters two significant challenges [4]. The large-scale sharing of vehicle data and the high privacy requirements of edge computing patterns are initially challenging to balance. The learning process is still vulnerable to modern adversarial techniques, such as data poisoning and model integrity attacks.

FL allows training a shared model on several edge devices without having to transfer raw data, which makes it a highly important approach to resolving data privacy problems. SFL collaborative learning architecture integrates the advantages of FL and Split Learning (SL). This architecture is characterized by client devices computing the first layers of the neural network, which they then send intermediate feature representations to a central server, which computes the rest of the forward pass. This design has the advantage of reducing computation load on edge clients, but has a greater communication overhead and may cause bottlenecks because training rounds can run sequentially. SFL provides a balanced trade-off by enabling clients to simultaneously train a sub-part of the model as in FL and at the same time partitioning the model between client and server, such as in SL. Such a structure facilitates the first-time identification and prevention of abnormal behaviors when they have not spread across the network [5], [6]. SFL improves processing and communication capabilities and thus can be used with scaled deployments in dynamic edge situations. The system design enables the sharing of models partially to complement responsiveness and reduce latency to ITS and other real-time applications [7]. Despite these benefits, current SFL systems continue to face limitations

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due to the temporal dynamics associated with transportation data and the development of adequate security mechanisms [8], [9], [10]. The quantum-inspired algorithmic frameworks and quantum-based security primitives of Edge-QSFL are significantly different. This framework utilizes quantum-inspired algorithmic principles to enhance traditional machine learning systems. Temperature-scaled aggregation approximates a Boltzmann distribution to produce perturbation-resistant client weighting. Quantum-based noise injection is employed to improve model generalization. Furthermore, Edge-QSFL employs quantum-resistant hashing primitives, such as SHA3-256, to ensure model integrity and quantum key distribution algorithms to preserve the communication channel between a client and a server. The concept uses quantum technology and advanced cryptography to strengthen standard computing infrastructure.

The proposed Edge-QSFL architecture integrates multi-scale temporal processing with quantum-enhanced security for intelligent transportation systems, addressing these critical challenges [11]. Quantum-secure protocols and multi-head TCNs are used to achieve complicated vehicular network traffic temporal patterns and privacy-preserving IDS. The quantum-inspired optimization promotes efficient learning across distributed edge nodes, while the adaptive aggregation approach strengthens resilience to cyber threats. The following are the primary contributions of the paper.

1) We introduce Edge-QSFL, a novel quantum-enhanced SFL framework that combines multi-head TCNs with quantum-enhanced ITS security measures. This integrated approach solves the important concerns of cybersecurity, privacy protection, and temporal pattern detection in distributed network traffic.
2) The proposed multi-head, quantum-enhanced TCN model is achieved by employing exponential dilation rates to produce a cascade of dilated causal convolutions with residual connections. This facilitates the retrieval of multi-scale time-varying features from periodic network-traffic measurements and preserves computational efficiency on resource-constrained edge computers.
3) We also provide a quantum-enhanced adaptive aggregation approach that leverages quantum-inspired temperature scaling to dynamically compensate client input based on pre-specified performance criteria. Model-poisoning risks are reduced, and the global model converges through client performance-based decisions.
4) We integrate robust quantum-security layers, including model-integrity checking and quantum key-generation techniques in distributed learning to defend against new attacks. The resulting security solution ensures verifiable implementation and the ability to model change throughout the federated network.

The remainder of this paper is organized as follows: Section II explains the related work, and Section III describes the proposed method. Section IV presents the experimental results, and Section V concludes the work.

II. Related Work

The deployment of edge-based FL for IDS significantly improves transportation security, offering an effective alternative to classical machine learning. This technique effectively overcomes two key issues associated with centralized approaches: data privacy concerns and the computing limits inherent in ITS [1]. Security measures are critically assessed as the industry moves from cloud-based solutions to edge computing, FL, and SFL-driven decentralized frameworks [3], [6], [7].

A. Edge Computing

Devarajan et al. [5] introduce security and efficiency in consumer IoV with blockchain and FL. Blockchain technology enables vehicle-wide model training and update consistency while protecting data. The system improves edge and cloud task offloading to improve traffic forecasting, security, and performance. Lin et al. [9] investigated edge computing IDS architecture and resource management. Their proposed approach effectively optimizes the distribution of various resources. They implement a Single-layer Dominant and Max-Min Fair (SDMMF) approach to ensure balance among various levels of a large-scale, edge-based IDS. They use Multilayer Dominant and Max-Min Fair (MDMMF) to manage system layer resources. Hasan et al. [12] suggested a federated learning technique to optimize computational offloading in 6G vehicle networks. This method ensures consistent performance across various hardware platforms while also improving privacy and security by training models locally on edge devices. The system updates vehicle information while optimizing 6G-V2X connectivity for self-driving capabilities. Cui et al. [13] introduce an improved Gradient Boosting Decision Tree model for secure edge computing in the Industrial Internet of Things (IIoT). Yao et al. [14] developed an edge-assisted technique to detect fraudulent attacks. Their wrapper-based feature selection improves IDS by removing irrelevant and redundant features. The algorithm divides attackers into four groups based on their capabilities and uses ensemble learning to improve detection rates. To reduce latency, several edge nodes execute the lightweight detection method simultaneously.

B. Federated Learning

Several studies [4], [5] utilized FL-based IDS techniques for transportation security. Mothukuri et al. [15] introduced an FL framework to enhance the security and privacy of IoT. Locally, it trains Gated Recurrent Unit (GRU) models on devices, transmitting only model updates to a central server rather than raw data. This solution protects user privacy while also creating a powerful global IDS. Dos Santos et al. [16] developed an FL system for network IDS that guarantees trustworthy model updates. Their system evaluates classification confidence independently, lowering model maintenance costs and errors. These techniques decreased false alarms by 12% missed detections by 9.6%, and modified the MAWIFlow dataset procedures. Ruzafa-Alc ázar et al. [17] investigated several privacy methodologies for training a federated IDS in IIoT situations. The authors compare traditional (FedAvg) and advanced (Fed+) aggregation methods on the non-uniform ToN_IoT dataset under various privacy requirements. The Fed+ algorithm retains detection accuracy even when privacy-protecting noise is used in training. Ullah et al. [18] proposed ZTID-IoV as a privacy-preserving IDS for IoVs. The system uses FL, a lightweight transformer, and meta-learning to identify potential risks without gathering user data. Neurosymbolic AI produces interpretable results with minimal processing resources, making it useful for resource-constrained devices in zero-trust contexts. De Oliveira et al. [19] proposed F-NIDS, which improves IDS scalability and data privacy. This system employs a federated AI architecture with asynchronous interactions to facilitate horizontal scaling and differential privacy to protect data integrity. The architecture is intended for flexible deployment in a variety of cloud or fog environments. The model accurately detects and classifies network threats while protecting privacy.

C. Split Federated Learning

Thapa et al. [20] introduced a method, SFL, that combines FL and SL. SFL solves the SL problem, protects data, and works with distributed learning models on resource-constrained devices. Furthermore, the SFL framework proposes using the idea of differential privacy to improve data security. Otoum et al. [21] proposed using an SL-based IDS for ITS to address security threats in wireless vehicle networks. The model employs SL to enhance the accuracy of detection and classification. SL enhanced FL and transfer learning models by 2-5% in accuracy and detection rates while conserving power. Turina et al. [22] propose SFL, a hybrid of federated and SL that incorporates their privacy and efficiency advantages. SFL has a superior accuracy-privacy tradeoff than parallel SL, allows for concurrent processing, and minimizes client processing requirements. Additionally, it maintains high accuracy on imbalanced data. The RingSFL is a distributed learning technique designed to overcome the fundamental limitations of standard FL [7]. RingSFL arranges clients in a ring topology and trains a subset of the model with FL and a model-splitting technique. This architecture solves the problem caused by device heterogeneity, resulting in significantly higher training efficiency. Local models are combined across the ring to prevent eavesdroppers from rebuilding the model or recovering raw training data. Pasquini et al. [23] explored potential vulnerabilities in split learning, a collaborative machine learning system. It shows that a malicious server can actively hijack learning to reassemble client training data. The developed threats are effective across a variety of datasets and can overcome newly proposed protective mechanisms. Furthermore, the study confirms that the protocol is vulnerable to attacks from malicious clients, introducing the known risks of FL to this decentralized system. Xu et al. [24] proposed an SFL system that combines FL parallel training with split learning model splitting. The method enables heterogeneous devices to individually select their optimal deep neural network split points, thereby addressing resource constraints in edge devices. To reduce system delay, it developed a strategy that optimizes both the selection of split points and the allocation of bandwidth simultaneously.

Our paper introduces Edge-QSFL, an innovative framework for IDS in ITS. The architecture combines quantum-secure SFL with multi-head temporal networks to develop a reliable and privacy-preserving IDS. It extracts features using lightweight TCNs on edge clients and aggregates them using quantum-inspired algorithms on a central server. This system exhibits high accuracy and increased resilience against cyber threats, providing a scalable and secure approach to transportation security.

III. Proposed Method: Edge-QSFL Framework

Figure 1 illustrates the Edge-QSFL framework, which provides a secure ITS solution. Quantum-enhanced learning and multi-scale temporal processing are integrated into a secure SFL system. The method leverages quantum principles to enhance resilience and efficiency. The complete architecture is detailed in the following subsections.

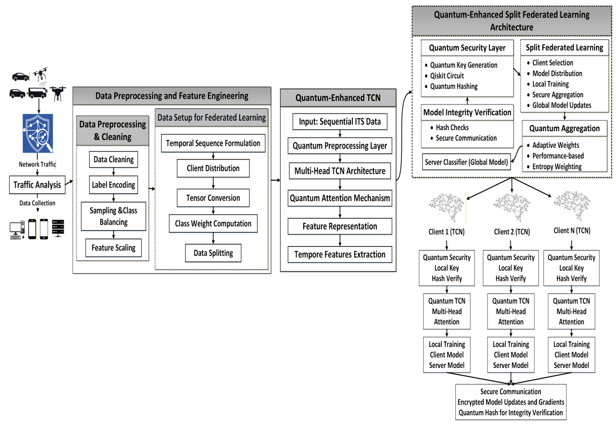


Fig. 1. Edge-QSFL: Quantum-enhanced split federated learning with multi-scale temporal processing for secure intelligent transportation networks.

A. Dataset Preparation

Three real-world datasets are used to evaluate the proposed approach. The CICIoV2024[1] [25] dataset aims to improve cybersecurity for IoVs. It includes authentic CAN bus data from a 2019 Ford vehicle, as well as five spoofing and Denial-of-Service (DoS) attacks. This dataset, designed as a benchmark, enables the development and evaluation of machine learning-based IDS to protect vehicles from cyber attacks. N-BaIoT[2] [26] is developed in a laboratory by attacking nine commercial IoT devices with Mirai botnet traffic, allowing assessment of the method and identifying malicious data from infected devices. The UAVAttackData[3] [27] dataset records GPS spoofing and jamming attacks on autonomous aerial vehicles (AAVs). It comprises data from both simulated and live flight testing, documenting both normal operations and attack scenarios. The live attacks are carried out utilizing a software-defined radio to broadcast fraudulent GPS locations or jamming signals. This dataset is a helpful resource for developing and assessing detection algorithms for typical AAV cybersecurity threats. Table I summarizes the class distribution for three cybersecurity datasets. The CICIoV2024 dataset for automobiles is strongly dominated by spoofing attacks (57.74%), including subcategories that break down individual attack types. UDP threats comprise 38.39% of the N-BaIoT dataset for IoT devices. The UAVAttackData datasets for drones are significantly dominated by benign traffic (77% and 86%), while jamming and spoofing attacks are the minority classes.

B. Quantum-Enhanced Data Preprocessing and Sequential Formulation

The quantum-enhanced preprocessing phase addresses class imbalances, where minority attack classes $\mathcal{C}_{\min}$ have significantly fewer samples than the main traffic class $\mathcal{C}_{\text{maj}}$.

[1] https://www.unb.ca/cic/datasets/iov-dataset-2024.html
[2] https://ieee-dataport.org/documents/n-baiot
[3] https://ieee-dataport.org/open-access/uav-attack-dataset

TABLE I
SUMMARY OF CLASS DISTRIBUTION ACROSS CICIOV2024, N-BAIOT, AND UAVATTACKDATA DATASETS

Dataset	Class	Samples (%)
CICIoV2024 (Main)	Benign	1,958 (19.58%)
	DoS	2,268 (22.68%)
	Spoofing	5,774 (57.74%)
CICIoV2024 (Sub)	DoS	2,908 (29.08%)
	GAS	1,855 (18.55%)
	RPM	1,539 (15.39%)
	Speed	1,831 (18.31%)
	Str. Wheel	1,867 (18.67%)
N-BaIoT	Ack	1,332 (13.32%)
	Scan	2,273 (22.73%)
	Syn	1,401 (14.01%)
	UDP	3,839 (38.39%)
	UDP Plain	1,155 (11.55%)
UAVAttackData (Jamming)	Benign	4,984 (77.34%)
	Jamming	1,460 (22.66%)
UAVAttackData (Spoofing)	Benign	3,123 (86.25%)
	Spoofing	498 (13.75%)

Furthermore, these procedures are adjusted to prepare the data for quantum enhancement and efficient IDS processing.

1) Data Cleaning and Balancing: We first employ detailed data cleaning to address missing values and infinite records in a raw dataset $\mathcal{D}_{\text{raw}} = \{(\mathbf{x}_i, y_i)\}_{i=1}^{N}$ with N samples, as illustrated in equation 1.

$$\mathcal{D}_{\text{clean}} = \{\mathbf{x}_i \in \mathcal{D}_{\text{raw}} \mid \mathbf{x}_i \neq \text{NaN} \wedge \mathbf{x}_i \neq \pm\infty\} \tag{1}$$

The Synthetic Minority Over-sampling Technique (SMOTE) with adaptive neighborhood parameters is used to achieve class balancing. Let $\mathcal{C}_k$ be the k-th class, which has $|\mathcal{C}_k|$ samples [28]. Using equation 2, the SMOTE algorithm produces synthetic samples for minority classes by linearly interpolating between a sample $\mathbf{x}_i$ and its k-nearest neighbors.

$$\mathbf{x}_{\text{new}} = \mathbf{x}_i + \lambda \cdot (\mathbf{x}_j - \mathbf{x}_i) \tag{2}$$

where $\lambda \sim \mathcal{U}(0, 1)$ is a random interpolation coefficient and $\mathbf{x}_j$ is a randomly chosen neighbor among the k-nearest neighbors of $\mathbf{x}_i$. Equation 3 illustrates the adaptive determination of the neighborhood size k. It ensures that small minority classes have valid neighborhood structure.

$$k = \min(5, |\mathcal{C}_{\min}| - 1) \tag{3}$$

2) Quantum-Inspired Feature Transformation: This includes amplitude encoding and a quantum-inspired transformation that projects features onto a unit sphere via L2 normalization (Eq. 4). This mimics quantum state representation, where $\|\tilde{\mathbf{x}}_i\|_2 = 1$ analogous to quantum state vectors. Unlike standard normalization, this creates a geometry compatible with subsequent quantum-inspired operations.

$$\tilde{\mathbf{x}}_i = \frac{\mathbf{x}_i}{\|\mathbf{x}_i\|_2} \tag{4}$$

Quantum computing data loading inspired this encoding paradigm, which offers three benefits. First, the angular separations of the unit sphere provide greater feature discrimination. Second, it is resistant to amplitude variance in IoT and IoV traffic feature sets; and third, it enables quantum-inspired noise injection strategies, making regularization easy.

$$\mathcal{D}_{\text{quantum}} = \left\{(\tilde{\mathbf{x}}_i, y_i) \mid \tilde{\mathbf{x}}_i = \frac{\mathbf{x}_i}{\|\mathbf{x}_i\|_2}, \forall (\mathbf{x}_i, y_i) \in \mathcal{D}_{\text{balanced}}\right\} \tag{5}$$

3) Sequential Data Formulation: The transformation of tabular data into sequential formulations is utilized to leverage the temporal features of cyber threats. The temporal sequences of length L are constructed as illustrated in equation 6, using a standardized feature matrix $\mathbf{X} \in \mathbb{R}^{N \times d}$.

$$\mathcal{S}_t = \{\mathbf{x}_{t-L+1}, \mathbf{x}_{t-L+2}, \ldots, \mathbf{x}_t\} \tag{6}$$

The sequence $\mathcal{S}_t$ represents temporal constraints. Equation 7 uses adaptive configuration to determine the sequence length L based on the application environment. Equation 8 is used to build the sequential dataset.

$$L = \begin{cases} L_{\text{long}} & \text{IoV CAN bus data multi-message patterns} \\ L_{\text{short}} & \text{real-time AAV threat detection} \\ L_{\text{medium}} & \text{IoT device traffic analysis} \end{cases} \tag{7}$$

$$\mathcal{D}_{\text{sequential}} = \{(\mathcal{S}_t, y_t)\}_{t=L}^{T} \tag{8}$$

The label y_t corresponds to the final element of the series $\mathcal{S}_t$, whereas T represents the total number of sequences. The sequence lengths are empirically set at $L_{\text{long}} = 10$ for CICIoV2024, $L_{\text{medium}} = 7$ for N-BaIoT, and $L_{\text{short}} = 5$ for UAVAttackData. These parameters emerged using systematic testing to balance temporal context capture and computational performance for each dataset domain.

4) Quantum Noise Injection for Robustness: We injected quantum-inspired noise during the training phase as a principled regularizer to enhance model robustness and generalization across diverse environments. We employ equation 9 to apply additive noise to each input sequence $\mathcal{S}_t$.

$$\mathcal{S}'_t = \mathcal{S}_t + \boldsymbol{\epsilon} \tag{9}$$

where $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2\mathbf{I})$ and the noise variance σ^2 is dynamically modified using equation 10 according to the features of the data set.

$$\sigma^2 = \alpha \cdot \text{Var}(\mathcal{D}_{\text{train}}) \cdot \beta_{\text{domain}} \tag{10}$$

A scaling factor is represented by α, and the variance of training features is Var($\mathcal{D}$train). For network traffic, the unpredictability of wireless channels is taken into consideration using a domain-specific multiplier, βdomain. The reliable representation of features is facilitated by this quantum-enhanced preprocessing. This preserves temporal relationships needed to detect cybersecurity risks in different applications.

C. Quantum-Optimized Multi-Head TCN Architecture

Figure 2 shows the architecture of the proposed quantum-enhanced TCN. A multi-head TCN with quantum-inspired techniques recognizes evolving attack patterns in the proposed model. These enhancements include consistent initialization, multi-scale feature capture, and adaptive attention. For stable and reliable FL convergence, the model uses a quantum-inspired approach to determine initial parameters (Eq. 12). Based on quantum superposition, the system analyzes attack data using exponential dilation rates ($d_h = 2^{h-1}$) across various time scales. Quantum-inspired attention mechanism (Eq. 13-14) dynamically adjusts network head priority, modeling quantum measurement probabilities to emphasize important temporal aspects for each input.

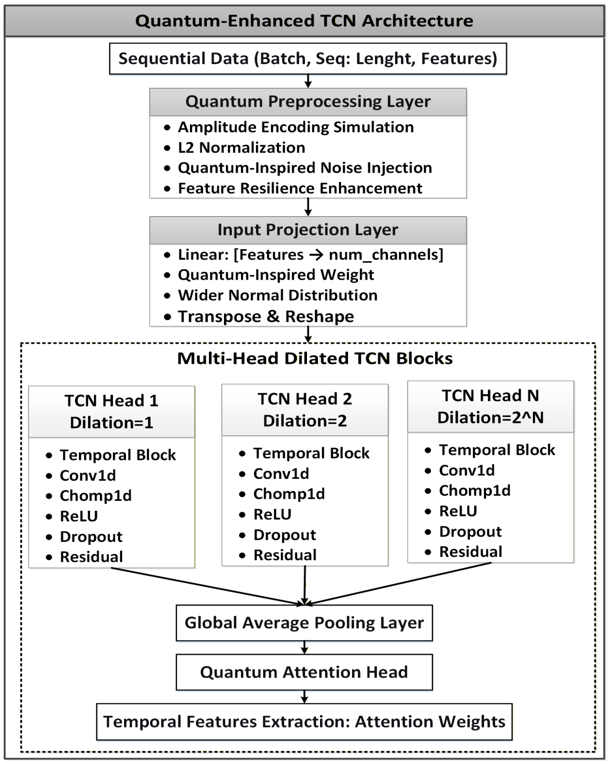


Fig. 2. Quantum-enhanced multi-head TCN architecture featuring dilated causal convolutions with residual connections for temporal feature extraction.

1) Multi-Scale Temporal Processing: The design includes H parallel TCN heads with exponentially increasing dilation rates $d_h = 2^{h-1}$, where $h = 1, 2, \ldots, H$. This multi-scale architecture allows for the simultaneous monitoring of both long-term intrusion patterns and short-term attack signatures [29], [30]. In Algorithm 1, each head processes input sequences using dilated causal convolutions and residual connections to maintain temporal causality and capture dependencies over extended time horizons. Equation 11 describes the temporal processing for each head h. The computational complexity of the client update in Algorithm 1 is $O(H \cdot L \cdot C^2)$, dominated by the parallel temporal convolutions across H heads for a sequence of length L and channel dimension C.

$$\mathbf{T}_h = \text{TemporalBlock}(\mathbf{Z}, d_h) \tag{11}$$

where $\mathbf{Z} = \mathbf{W}_{\text{proj}}\text{reshape}(\mathbf{X}')$ represents the expected input sequence. The $\mathbf{T}_h \in \mathbb{R}^{B \times C \times L}$ indicates the temporal features retrieved by head h, where B is the batch size, C is the number of channels, and L is the sequence length.

2) Quantum-Inspired Parameter Initialization: A modified normal distribution derived from the probability densities of quantum harmonic generators is used to initialize the model parameters in a quantum-inspired approach. To initialize each weight matrix $\mathbf{W} \in \mathbb{R}^{m \times n}$, use equation 12.

$$\mathbf{W}_{ij} \sim \mathcal{N}\left(0, \sigma^2 \cdot \sqrt{\frac{2}{m+n}}\right) \tag{12}$$

where the variance scaling is improved by the principles of quantum uncertainty, with $\sigma > 1$. Convergence stability is improved across diverse client data distributions, and gradient flow is enhanced during federated training using this initialization method.

Algorithm 1 Quantum-Enhanced Client Update With Multi-Head TCN Processing and Security Verification

Input: Client k with data $\mathcal{D}_k$, global parameters Θ_c, Θ_s, sequence length L, heads H, quantum security flag Q
Output: Updated parameters $\Theta_c^{(k)}, \Theta_s^{(k)}$, attention weights A, quantum hashes $\mathcal{H}$, client metrics $(acc_k, loss_k)$

Initialization: $\Theta_c^{(k)} \leftarrow \Theta_c$, $\Theta_s^{(k)} \leftarrow \Theta_s$;
if $Q = \mathit{True}$ **then**
  $K_q \leftarrow \text{QuantumKeyGeneration}()$
$\mathcal{B} \leftarrow \text{SequenceBatch}(\mathcal{D}_k, L)$;
Quantum-Enhanced Training:;
for *each batch* $(X_b, y_b) \in \mathcal{B}$ **do**
  if $Q = \mathit{True}$ **then**
    $X_b' \leftarrow \text{QuantumPreprocessing}(X_b)$
  Multi-Head TCN Processing:;
  $Z \leftarrow W_{proj}\text{reshape}(X_b')$; `// Input projection`
  $\mathcal{H} \leftarrow \{\}$;
  for $h = 1$ **to** H **do**
    $T_h \leftarrow \text{TemporalBlock}(Z, 2^{h-1})$; `// Dilated convolution`
    $h_h \leftarrow \text{GlobalAvgPool}(T_h)$;
    $\mathcal{H} \leftarrow \mathcal{H} \cup \{h_h\}$;
  Attention & Classification:;
  $\alpha \leftarrow \text{softmax}(W_2 \tanh(W_1 \text{concat}(\mathcal{H})))$; `// Attention weights`
  $h_{final} \leftarrow \sum_{h=1}^{H} \alpha_h \cdot h_h$;
  $\hat{y}_b \leftarrow f_s(h_{final}; \Theta_s^{(k)})$;
  Optimization:;
  $\mathcal{L} \leftarrow \text{CrossEntropy}(\hat{y}_b, y_b) + \lambda\|\Theta\|_2^2$;
  $\Theta_c^{(k)} \leftarrow \Theta_c^{(k)} - \eta\nabla_{\Theta_c^{(k)}}\mathcal{L}$;
  $\Theta_s^{(k)} \leftarrow \Theta_s^{(k)} - \eta\nabla_{\Theta_s^{(k)}}\mathcal{L}$;
Security & Evaluation:;
if $Q = \mathit{True}$ **then**
  $\mathcal{H} \leftarrow \{\text{QuantumHash}(\Theta_c^{(k)}), \text{QuantumHash}(\Theta_s^{(k)})\}$
$acc_k, loss_k \leftarrow \text{Evaluate}(\Theta_c^{(k)}, \Theta_s^{(k)}, \mathcal{D}_k)$;
return $\Theta_c^{(k)}, \Theta_s^{(k)}, \alpha, \mathcal{H}, (acc_k, loss_k)$;

3) Attention-Based Feature Fusion: A quantum-inspired attention mechanism combines the outputs from several TCN heads by dynamically weighting the contributions of each head. The compressed temporal features from head h are represented by $\mathbf{h}_h = \text{GlobalAvgPool}(\mathbf{T}_h)$. The attention weights are calculated using the equation 13.

$$\alpha_h = \frac{\exp(\mathbf{W}_2 \tanh(\mathbf{W}_1 \mathbf{h}_h))}{\sum_{j=1}^{H} \exp(\mathbf{W}_2 \tanh(\mathbf{W}_1 \mathbf{h}_j))} \tag{13}$$

where $\mathbf{W}_1 \in \mathbb{R}^{d_a \times d_h}$ and $\mathbf{W}_2 \in \mathbb{R}^{1 \times d_a}$ are learnable projection matrices, and d_a is the attention dimension. The final attended representation is obtained through equation 14.

$$\mathbf{h}_{\text{final}} = \sum_{h=1}^{H} \alpha_h \cdot \mathbf{h}_h \tag{14}$$

The model uses adaptive integration to emphasize relevant temporal scales for individual attack types, such as short-term patterns for jamming detection and long-term dependencies for complex spoofing attacks.

4) Temporal Block Design: The equation 15 is employed by each TemporalBlock to implement dilated causal convolutions with gated activation functions.

$$\text{TemporalBlock}(\mathbf{X}, d) = \text{ReLU}(\mathbf{X} + \text{Conv1D}_d(\mathbf{X})) \tag{15}$$

To ensure that the output at time t depends only on inputs up to time t, the dilated causal convolution with a dilation rate d is denoted as Conv1D_d. Equation 16 summarizes the full forward pass of the TCN that has been improved for quantum computing.

$$\mathbf{h}_{\text{final}}, \boldsymbol{\alpha} = f_c(\mathbf{X}; \Theta_c) \tag{16}$$

The attention weights across all heads are represented by $\boldsymbol{\alpha} = [\alpha_1, \alpha_2, \ldots, \alpha_H]$, and the client-side model parameters are denoted by Θ_c. This architecture establishes a strong foundation for the extraction of temporal features.

D. Quantum Security Layer Implementation

We employ a comprehensive quantum security layer to address critical security requirements in FL environments, ensuring end-to-end protection throughout the learning process.

1) Quantum Key Distribution Simulation: Secure client-server communication is achieved through the simulation of quantum key distribution principles [10]. The quantum circuit described in Equation 17 is simulated within the Qiskit environment, with Hadamard gates establishing superposition and CNOT gates implementing qubit entanglement.

$$|\psi\rangle = \bigotimes_{i=1}^{N_q} H|0\rangle, \quad U_{\text{ent}} = \prod_{i=1}^{N_q-1} \text{CNOT}(i, i+1) \tag{17}$$

The quantum circuit simulation is executed using Qiskit's Aer simulator, producing measurement outcomes that form cryptographically secure keys. In practice, this simulation runs on classical hardware and serves as a quantum-inspired key generation mechanism. These keys are then used for establishing encrypted channels for transmitting intermediate features $\mathbf{h}_k$ and seeding the quantum-resistant hashing in the integrity verification layer.

2) Model Integrity Verification: Each client calculates quantum-resistant hash values of local model parameters using the SHA3-256 algorithm, as shown in equation 18:

$$\mathcal{H}_q = \text{SHA3-256}(\Theta^{(k)}) \tag{18}$$

These cryptographic hashes serve as digital fingerprints during federated aggregation to detect model manipulation or Byzantine attacks.

3) Quantum Entropy Monitoring: We continuously monitor quantum-inspired entropy throughout federated learning rounds, using equation 20:

$$\mathcal{E}_t = -\sum_{k=1}^{|\mathcal{S}_t|} w_k^{(t)} \log(w_k^{(t)}) \tag{19}$$

where $\mathcal{E}_t$ denotes quantum-inspired entropy at round t, and $w_k^{(t)}$ are aggregation weights. Significant changes in entropy patterns indicate potential adversarial activity.

4) Quantum Entropy Monitoring: The vulnerabilities can be detected by continuously monitoring quantum entropy measures throughout federated learning rounds, utilizing equation 20.

$$\mathcal{E}_t = -\sum_{k=1}^{|\mathcal{S}_t|} w_k^{(t)} \log(w_k^{(t)}) \tag{20}$$

The quantum entropy at round t is denoted by $\mathcal{E}_t$, and the aggregation weights are $w_k^{(t)}$. Significant changes in entropy patterns warn security personnel of probable adversary activity. The privacy and integrity of the FL process are preserved, while effective defense against a variety of threats is ensured by this multi-layered security approach.

E. Split Federated Learning Framework

Our approach employs an SFL architecture to distribute the quantum-optimized TCN between edge clients and a central server [6], [7], balancing privacy preservation and detection performance as shown in Algorithm 2. The computation per-round cost of the Edge-QSFL coordination method described in Algorithm 2 is of the order of $O(K \cdot (H \cdot L \cdot C^2) + K \log K)$, where K is the size of the client group. The first term corresponds to K client updates, each with the complexity $O(H \cdot L \cdot C^2)$ as analyzed in Algorithm 1. The second term accounts for the overhead of quantum-enhanced, adaptive aggregation.

1) Client-Server Model Partitioning: The model is split at the final attention-pooling layer of the multi-head TCN. Clients retain the temporal processing layers to locally analyze raw sensor data, producing only the intermediate feature vector $\mathbf{h}_k = \mathbf{h}_{\text{final}} \in \mathbb{R}^{d_h}$ with $d_h = 128$ using equation 21.

$$\mathbf{h}_k, \boldsymbol{\alpha} = f_c(\mathbf{X}_k; \Theta_c). \tag{21}$$

Transmitting this compressed, abstract representation $\mathbf{h}_k$, rather than raw data $\mathbf{X}_k$ or full gradients, enhances privacy: the non-linear features are difficult to invert, significantly reducing data leakage risks compared to standard FL.

2) Server-Side Classification: The server hosts the classification head, which processes the received features to generate final predictions using equation 22.

$$\hat{\mathbf{y}} = f_s(\mathbf{h}_k; \Theta_s). \tag{22}$$

This split design reduces communication overhead while maintaining privacy across all application domains [8], [20].

3) Federated Learning Formulation: The joint optimization objective combines client-side feature extraction and server-side classification using equation 23.

$$\min_{\Theta_c, \Theta_s} \sum_{k=1}^{N} \mathbb{E}_{(\mathbf{X},\mathbf{y})\sim\mathcal{D}_k}[\mathcal{L}(f_s(f_c(\mathbf{X}; \Theta_c); \Theta_s), \mathbf{y})], \tag{23}$$

where $\mathcal{L}$ is the cross-entropy loss and $\mathcal{D}_k$ is the local data distribution at client k.

Algorithm 2 Edge-QSFL: Quantum-Enhanced SFL for Secure ITS

Input: IoV dataset $\mathcal{D}$, clients N, rounds T, sequence length L, heads H, quantum enhancement Q, participation rate p, input dimension d_{in}, hidden dimension d_{hidden}
Output: Global parameters Θ_c^*, Θ_s^*

```
Phase 1: Quantum-Enhanced Data Preprocessing;
D_clean ← RemoveNaN(D);
D_balanced ← SMOTE(D_clean);
x̃_i ← (x_i − μ)/σ ;                    // Standardization
D_train, D_test ← TrainTestSplit(D_balanced);
{D_k}_{k=1}^N ← IIDPartition(D_train, N);
Phase 2: Quantum-Enhanced Model Initialization;
f_c(·; Θ_c) ← QuantumOptimizedTCN(d_in, H, L, Q) ;
  // Client-side TCN: d_in is input feature dimension
f_s(·; Θ_s) ← ServerSideClassifier(H × d_hidden, Q) ;
  // Server-side classifier: d_hidden is per-head feature dimension
Q ← QuantumSecurityLayer() ;           // Quantum security layer
Phase 3: Quantum-Enhanced Federated Training;
for t = 1 to T do
    S_t ⊂ {1, ..., N} with |S_t| = ⌊pN⌋ ; // Select participants
    U_c ← {}, U_s ← {}, A ← {};
    for each client k ∈ S_t in parallel do
        Θ_c^(k), Θ_s^(k), α, H_q, (acc_k, loss_k) ←
          Algorithm 1(k, Θ_c, Θ_s, L, H, Q);
        U_c ← U_c ∪ {Θ_c^(k)};
        U_s ← U_s ∪ {Θ_s^(k)};
        A ← A ∪ {acc_k};
    Quantum-Enhanced Adaptive Aggregation:;
    τ ← max(0.1, 0.5 − σ(A) × 0.3) ;  // Quantum temperature scaling
    w_k = exp(acc_k/τ) / Σ_{j∈S_t} exp(acc_j/τ) ;
      // Performance-based weights
    E ← −Σ w_k log(w_k) ; // Quantum entropy for diversity
    Model Aggregation:;
    Θ_c ← Σ_{k∈S_t} w_k · Θ_c^(k) ;       // Aggregate client models
    Θ_s ← Σ_{k∈S_t} w_k · Θ_s^(k) ;       // Aggregate server models
    Global Model Evaluation:;
    loss_g, acc_g, f1_w, f1_m ←
      EvaluateGlobal(Θ_c, Θ_s, D_test);
Phase 4: Quantum Security Verification;
if Q = True then
    H_final ← QuantumHash(Θ_c, Θ_s);
    VerifyModelIntegrity(H_final);
return Θ_c, Θ_s;
```

4) Dynamic Client Participation: The participation rate p can be changed using equation 24, enabling the system to accommodate adjustable participation patterns. This enables intermittent connectivity in resource-constrained edge devices, ensuring dependable functioning in real-world deployment scenarios.

$$|\mathcal{S}_t| = \lfloor pN \rfloor \tag{24}$$

F. Quantum-Inspired Adaptive Aggregation Mechanism

The aggregation phase uses a novel quantum-inspired weighting technique to dynamically adjust client contributions based on data quality and performance. To compute aggregation weights, we employ a quantum Boltzmann distribution, with client accuracies treated as energy levels rather than uniform averaging. Equation 25 scales performance metrics exponentially to derive client aggregation weights.

$$w_k = \frac{\exp(acc_k/\tau)}{\sum_{j\in\mathcal{S}_t}\exp(acc_j/\tau)} \tag{25}$$

where acc_k represents client k's validation accuracy, and τ is a quantum-inspired temperature parameter that controls selectivity. The temperature adapts based on performance variance across clients using equation 26.

$$\tau = \max(\tau_{\min}, \tau_{\max} - \gamma \cdot \sigma(\mathcal{A})) \tag{26}$$

The standard deviation of client accuracy is $\sigma(\mathcal{A})$, temperature boundaries are $\tau_{\min}$ and $\tau_{\max}$, and scaling factor is γ. This adaptive temperature allows stable aggregation during the initial learning phases, but becomes more selective as models converge. Equation 27 is employed to aggregate client-side parameters in the model. Similarly, server-side parameters $\Theta_s^{(t+1)}$ are aggregated. Quantum entropy is monitored constantly by the framework to ensure model diversity, as indicated by equation 28.

$$\Theta_c^{(t+1)} = \sum_{k\in\mathcal{S}_t} w_k \cdot \Theta_c^{(k)} \tag{27}$$

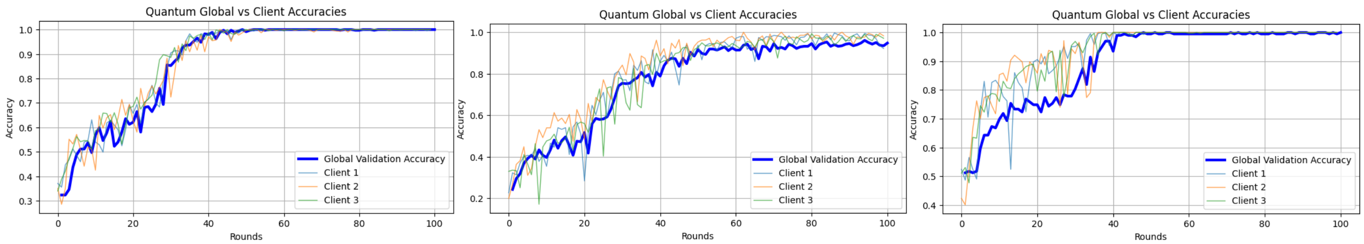
$$\mathcal{E}_t = -\sum_{k=1}^{|\mathcal{S}_t|} w_k \log(w_k) \tag{28}$$

This entropy metric functions as a diversity indicator, ensuring that the model maintains beneficial variation during training while simultaneously preventing adverse divergence. The aggregation approach easily handles heterogeneous data distributions across various vehicle models, IoT device types, and AAV systems, while prioritizing clients for improved data quality and better model performance.

IV. Results and Discussions

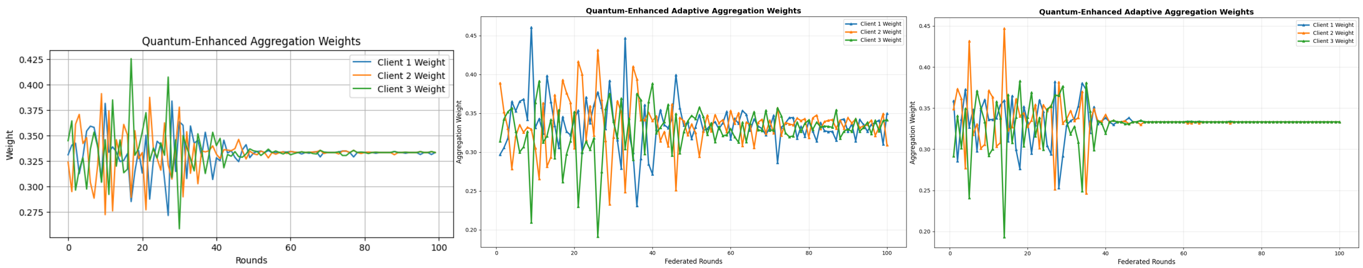
A. Performance Metrics

We chose a variety of network traffic-based datasets and several clients to thoroughly test the proposed method. The suggested method is tested with three publicly available standard datasets: CICIoV2024, N-BaIoT, and UAVAttackData. We evaluated performance by utilizing precision, recall, F-measure, and accuracy, respectively. These measures are derived from True Positive (TP), False Positive



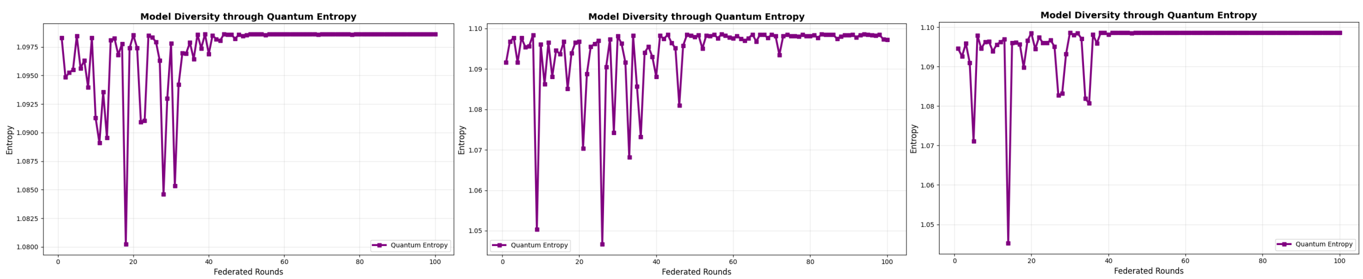
(a) CICIoV2024 (b) N-BaIoT (c) UAVAttackData

Fig. 3. Accuracy progression of individual clients and the global model during federated training with 3 clients, demonstrating stable convergence across the (a) CICIoV2024, (b) N-BaIoT, and (c) UAVAttackData datasets.



(a) CICIoV2024 (b) N-BaIoT (c) UAVAttackData

Fig. 4. Evolution of quantum-enhanced aggregation weights across training rounds with 3 clients, showing dynamic prioritization of reliable contributors to mitigate poisoning attacks on the (a) CICIoV2024, (b) N-BaIoT, and (c) UAVAttackData datasets.



(a) CICIoV2024 (b) N-BaIoT (c) UAVAttackData

Fig. 5. Temporal progression of quantum entropy during aggregation with 3 clients, showing stable values that indicate balanced participation and prevent client dominance across the (a) CICIoV2024, (b) N-BaIoT, and (c) UAVAttackData datasets.

(FP), True Negative (TN), and False Negative (FN) values. Equation 29-32 provide performance measurements.

$$\text{Precision} = \frac{TP}{TP + FP} \tag{29}$$

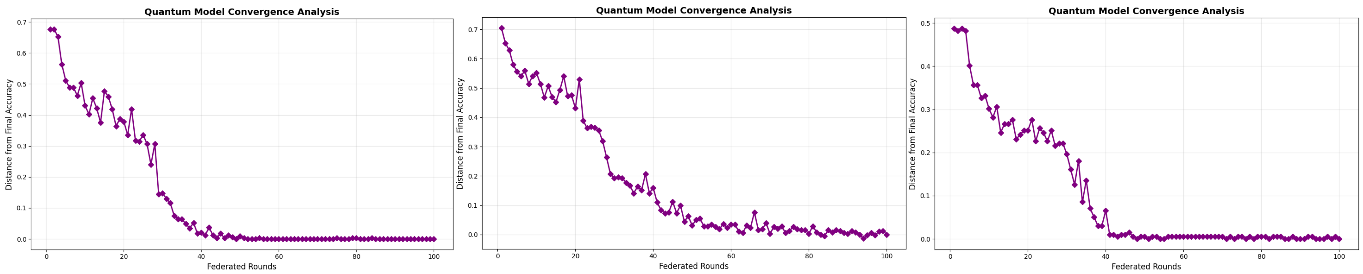
$$\text{Recall} = \frac{TP}{TP + FN} \tag{30}$$

$$F\text{-measure} = \frac{2 \times TP}{2 \times TP + FP + FN} \tag{31}$$

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{32}$$

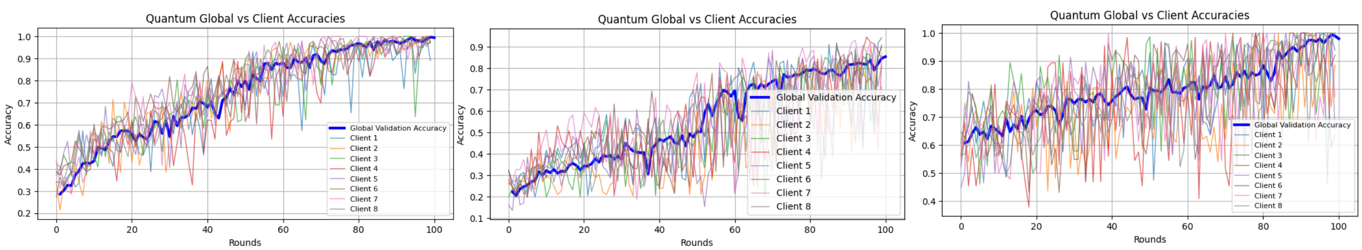
B. Performance Analysis and Comparisons

Figure 3 shows that the quantum-enhanced FL model showed consistent performance gains across all three datasets across 100 training iterations. Accuracy in the CICIoV2024 dataset improved from a range of 0.3362–0.3821 to 1.0. The accuracy of the N-BaIoT dataset increased from 0.1994–0.3299 to a range of 0.9217–0.9824 across five IoT attack categories. During training, all datasets demonstrated considerable accuracy gains, with the greatest peaks of 0.9653-0.9870, 0.8885-0.9326, and 0.9547-0.9831, respectively. Figure 4 illustrates the progression of the aggregation weights with each subsequent federated round. client 1 scores 0.2852 and client 3 scores 0.4254 at round 18, and the two scores converge to approximately 0.333 by round 30 in the CICIoV2024 case. This resulted in a 32.37% accuracy shift to complete 100% success, which is sustained by a stable quantum entropy of 1.098. A balance of 0.333 and an entropy of 1.097 in the five-class experiment adjusted accuracy from 24.28% to 94.78%. Similar to UAVAttackData, quantum-enhanced aggregation is stable and versatile across smart-city purposes. Figure 5 illustrates quantum-enhanced entropy monitoring results. For the CICIoV2024 dataset, entropy remained high and stable over 50 rounds, averaging 1.098. For the N-BaIoT dataset, entropy values between 2.07 and 2.08 over 100 rounds indicated excellent diversity and balanced client participation, supporting the final 99.42% accuracy. Figure 6 depicts the model's convergence. On CICIoV2024, loss dropped from 1.1871 to near zero over 100 rounds, improving accuracy from 32.37% to 100%. Similar stable convergence was observed on other datasets. Figure 7 shows the accuracy trend. For CICIoV2024, accuracy surged from 28.61% to 99.42%,



(a) CICIoV2024 (b) N-BaIoT (c) UAVAttackData

Fig. 6. Global training loss reduction over federated rounds with 3 clients, demonstrating smooth convergence and validating the stability of the quantum-inspired aggregation across the (a) CICIoV2024, (b) N-BaIoT, and (c) UAVAttackData datasets.



(a) CICIoV2024 (b) N-BaIoT (c) UAVAttackData

Fig. 7. Accuracy progression with 8 clients, demonstrating maintained convergence and performance scalability with a larger, heterogeneous client pool across the (a) CICIoV2024, (b) N-BaIoT, and (c) UAVAttackData datasets.

TABLE II
PERFORMANCE RESULTS FOR CICIOV2024 DATASET WITH DIFFERENT NUMBERS OF CLIENTS

Number of Clients	Class	Precision	Recall	F1-score
3 Clients	Benign	1.00	0.88	0.93
	DoS	1.00	1.00	1.00
	Spoofing	0.88	1.00	0.93
8 Clients	Benign	1.00	0.98	0.99
	DoS	1.00	1.00	1.00
	Spoofing	0.98	1.00	0.99
14 Clients	Benign	0.76	0.66	0.71
	DoS	0.89	0.89	0.89
	Spoofing	0.69	0.80	0.74
14 Clients (Main)	DoS	0.93	1.00	0.96
	GAS	0.94	1.00	0.97
	RPM	0.95	0.98	0.97
	Speed	1.00	1.00	1.00
	Str. Wheel	1.00	0.80	0.89

TABLE III
PERFORMANCE RESULTS FOR N-BAIOT DATASET WITH DIFFERENT NUMBERS OF CLIENTS

No. of Clients	Class	Precision	Recall	F1-score
3 Clients	Ack	0.87	0.85	0.86
	Scan	1.00	0.99	0.99
	Syn	0.98	1.00	0.99
	UDP	0.96	0.92	0.94
	UDP Plain	0.91	0.96	0.94
8 Clients	Ack	0.67	0.68	0.67
	Scan	0.99	1.00	0.99
	Syn	0.99	0.99	0.99
	UDP	0.78	0.86	0.82
	UDP Plain	0.81	0.71	0.74
13 Clients	Ack	0.43	0.66	0.52
	Scan	0.91	0.99	0.95
	Syn	0.96	0.89	0.92
	UDP	0.77	0.52	0.62
	UDP Plain	0.75	0.67	0.71

reaching 50% by round 11 and 85% by round 40. The N-BaIoT and UAVAttackData datasets exhibited similar adaptive and stable learning across smart city environments.

The performance evaluation of the model across three datasets is summarized in Tables II to IV. On the CICIoV2024 dataset, Table II shows the model achieves near-perfect detection for DoS and spoofing attacks with 3 and 8 clients. However, scaling to 14 clients degrades performance for certain classes, such as the benign class, where the F1-score drops to 0.71, though configuration remains critical as the Speed class retains perfect detection. Table III details results for the N-BaIoT dataset, where high initial F1-scores, such as 0.99 for Scan and Syn classes, are achieved with 3 clients. Performance declines as clients increase, exemplified by the Ack class F1-score falling from 0.86 to 0.52 when scaling from 3 to 13 clients. For the UAVAttackData dataset in Table IV, performance varies by attack type. Jamming detection reaches

TABLE IV
PERFORMANCE RESULTS FOR UAVATTACKDATA DATASET WITH DIFFERENT NUMBERS OF CLIENTS AND ATTACK TYPES

Number of Clients	Class	Precision	Recall	F1-score
8	Benign	0.99	0.97	0.98
	Jamming	0.97	0.99	0.98
13	Benign	0.69	0.77	0.73
	Jamming	0.73	0.64	0.68
3	Benign	1.00	0.84	0.91
	Spoofing	0.86	1.00	0.93
8	Benign	0.66	0.98	0.79
	Spoofing	0.97	0.49	0.65
12	Benign	0.88	0.74	0.81
	Spoofing	0.78	0.90	0.84

an F1-score of 0.98 with 8 clients, falling to 0.68 with 13 clients. Spoofing detection shows variability, scoring 0.93, 0.65, and 0.84 with 3, 8, and 12 clients respectively. These

TABLE V
PERFORMANCE COMPARISON ACROSS DIFFERENT CLIENT CONFIGURATIONS USING CICIOV2024 DATASET

Clients	Accuracy	Loss	F1 Weighted	F1 Macro	Client Accuracies
3	0.960	0.109	0.960	0.956	0.946, 0.987, 0.982
8	0.994	0.031	0.994	0.994	0.890, 0.980, 0.970, 0.974, 0.990, 1.000, 1.000, 0.978
14	0.789	0.492	0.788	0.780	0.889, 0.794, 0.788, 0.734, 0.945, 0.838, 0.683, 0.741, 0.667, 0.620, 0.894, 0.674, 0.950, 0.856
14	0.962	0.141	0.961	0.957	0.711, 0.818, 0.887, 0.546, 0.902, 0.965, 0.604, 0.939, 0.911, 0.810, 0.897, 1.000, 1.000, 0.729

TABLE VI
PERFORMANCE COMPARISON ACROSS DIFFERENT CLIENT CONFIGURATIONS USING N-BAIOT DATASET

Clients	Accuracy	Loss	F1 Weighted	F1 Macro	Client Accuracies
3	0.948	0.213	0.948	0.944	0.922, 0.982, 0.971
8	0.854	0.378	0.853	0.845	0.943, 0.810, 0.898, 0.844, 0.821, 0.908, 0.872, 0.713
13	0.749	0.569	0.754	0.744	0.686, 0.817, 0.750, 0.337, 0.765, 0.962, 0.795, 0.522, 0.661, 0.825, 0.793, 0.510, 0.811

TABLE VII
PERFORMANCE COMPARISON ACROSS DIFFERENT CLIENT CONFIGURATIONS USING UAVATTACKDATA DATASET

Clients	Accuracy	Loss	F1 Weighted	F1 Macro	Client Accuracies
8 (Jamming)	0.980	0.057	0.980	0.980	0.939, 0.885, 1.000, 0.636, 0.655, 0.922, 0.773, 1.000
3 (Spoofing)	0.920	0.323	0.919	0.919	0.608, 0.928, 1.000
8 (Spoofing)	0.736	0.930	0.719	0.720	0.613, 0.722, 0.447, 0.707, 0.944, 0.812, 1.000, 0.883
12 (Spoofing)	0.824	0.502	0.823	0.823	0.927, 0.684, 0.571, 0.531, 0.759, 0.963, 0.917, 0.409, 0.950, 0.737, 0.471, 0.885

TABLE VIII
PERFORMANCE COMPARISON WITH RELATED PUBLISHED STUDIES

Work	Method	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
Jinglong et al. [7]	Adaptive SFL	99.10	–	–	–
Otoum et al. [21]	Transfer learning with SFL	98.50	–	–	–
Latif et al. [31]	HMAC with SFL	98.06	99.53	100	99.76
Lu et al. [32]	Self-Labeled Personalized FL	98.1	92.1	88.1	88.9
Proposed Approach	**Edge-QSFL**	**99.42**	**100**	**99.26**	**99.37**

TABLE IX
ABLATION STUDY: COMPARISON WITH BASELINE METHODS

Method / Variant	Key Characteristics	Accuracy (%)	Conv. Rnd.	Comm. (MB)
Standard FL (FedAvg)	Full model, uniform averaging	94.7	66	15.4
SFL-AdaptSplit	Dynamic model partitioning	96.7	46	8.6
Full Edge-QSFL	**Multi-head TCN + QE-AA**	**99.4**	**39**	**8.5**
Edge-QSFL (FedAvg Agg)	Multi-head TCN + Std. Agg	97.5	44	8.5

results indicate that while adding clients generally reduces performance, the specific impact depends on the attack type.

The overall performance of the model on the CICIoV2024 dataset under various client settings is summarized in Table V. The model obtains an accuracy of 0.994 with 8 clients under optimal conditions. Table VI shows that performance decreases with increasing client count, with an accuracy of 0.749 for 13 clients. A significant experiment with 14 clients showed good accuracy, such as 0.962, proving that data distribution is more important than client count. Table VII shows that performance is high for jamming accuracy 0.98 but lowers for spoofing, such as 0.736, with only altered setups restoring performance. These results show a fundamental scalability trade-off: client engagement increases noise and heterogeneity, which degrades the global model. As a result, system designers must prioritize data quality and training dynamics over solely aggregating additional clients. These results show that attack type and client distribution greatly affect classification accuracy.

Table VIII compares the proposed Edge-QSFL system to related studies. Edge-QSFL attains the highest overall performance, with a 99.37% F1-score, 100%, 99.26% recall, and 99.42% accuracy. The proposed framework outperforms the related studies in an edge-based learning environment, proving its dependability and effectiveness. Table IX presents an ablation study. It compares Edge-QSFL with standard baselines and a key variant where only the aggregation mechanism is changed. The results show that while the split-learning architecture (SFL-AdaptSplit) improves over standard FL, our proposed method, Quantum-Enhanced Adaptive Aggregation (QE-AA) is the critical component. Replacing QE-AA with standard averaging causes a clear drop in accuracy (from 99.4% to 97.5%) and slower convergence (44 rounds vs. 39).

V. CONCLUSION

The Edge-QSFL work integrates a multi-head TCN and quantum-inspired safety measures to enhance smart transportation infrastructures. This design can efficiently utilize temporal network data while maintaining privacy with a decentralized learning regime that integrates edge devices and central servers. The adaptive aggregation protocol, supported by quantum-inspired temperature modulation, protects against data poisoning and the changing spectrum of cyber threats. The quantum-enhanced multi-head TCN architecture captures sequential traffic by utilizing multi-scale features. Embedding quantum security layers, using quantum-resistant hashing, and modeling quantum key distribution show model consistency. The approach performed well empirically across three different real-world datasets. Edge-QSFL detected vehicle-specific threats, including CAN bus spoofing and DoS, with 99.42% accuracy at CICIoV2024. N-BaIoT successfully classified Mirai botnet variants with 94.82% accuracy for IoT device security. In AAV scenarios (UAVAttackData), it achieved an accuracy of 98.0% in distinguishing benign from GPS interference and spoofing. Quantum-enhanced adaptive aggregation preserved high client diversity and proactively prioritized trusted contributors, improving resilience to harmful updates.

Edge-QSFL demonstrated satisfactory performance with 3–8 clients; however, we observed a scalability trade-off, as accuracy decreased as client pools expanded (e.g., at 14 clients). This suggests that hierarchical aggregation or client clustering may be advantageous in environments that are exceedingly heterogeneous. Quantum cryptography, such as realistic quantum key distribution, may be explored to improve model update security. However, processing, memory, energy, and quantum hardware restrictions make quantum cryptography on edge devices difficult. Lightweight quantum-compatible hardware interfaces or efficient hybrid classical–quantum protocols are needed to address these issues. The quantum-inspired components can be optimized for resource-constrained edge devices and integrate client selection and hierarchical aggregation algorithms to enable larger and more complicated transportation networks.

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